A Self-Supervised Learning Pretraining Framework for Remote Sensing Image Change Detection

Ling Wan, Yuming Xiang, *Senior Member, IEEE*, Wenchao Kang, and Lei Ma

***Abstract*— In recent years, change detection (CD) has achieved remarkable success through the use of deep learning. However, most existing methods rely on label teaching and thus have many limitations when dealing with the complexity and diversity of remote sensing (RS) scenes. In this work, we propose a self-supervised learning (SSL) pretraining framework for RS image CD (SSLCD). Our motivation is to leverage the intrinsic structure of multitemporal data to learn general and robust representations, generating competitive pretrained models for the CD task. On the one hand, an intrinsic structure learning (ISL) strategy is introduced, which enforces feature invariance and change consistency across temporal phases and augmented views, learning discriminatory representations related to changes while simultaneously mitigating noises associated with irrelevant changes. On the other hand, a self/cross-reconstruction (SCR) mechanism is proposed, which extends masked image modeling (MIM) to multitemporal images by predicting missing parts of the pre-phase image using the post-phase image, thereby enhancing the model's capacity in modeling high-level contextual information. Finally, models pretrained using SSLCD are extensively evaluated on three CD datasets, and the results demonstrate that SSLCD outperforms existing RS pretraining methods and the state-of-the-art CD methods.**

***Index Terms*— Change detection (CD), intrinsic structure learning (ISL), self/cross-reconstruction (SCR), self-supervised learning (SSL).**

I. Introduction

As REMOTE sensing (RS) technology advances, numerous images provide comprehensive earth observation. Faced with a proliferation of RS images, change detection (CD) task is more meaningful by obtaining the changed information, especially in the fields of land monitoring, disaster reduction, and early warning [1].

Generally, a CD method inputs multitemporal images and analyzes spectral or class differences to predict "where" the change occurs [2]. Representative methods include the arithmetic operations-based methods [3], [4], transformation-based methods [5], [6] and classification-based methods [7]. However, these traditional methods, relying on empirical design, often cannot meet the accuracy requirements of high-resolution RS images. Recently, deep learning has demonstrated promising potential in CD tasks, evolving from fully convolutional networks (FCNs) to convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) [8], graph convolutional networks (GCNs) [9], Transformer [10], [11], and diffusion models [12]. However, the above deep learning-based CD methods rely on the teaching of labels and thus have many limitations when dealing with the complexity and diversity of RS scenes.

1) *CD is application-oriented:* For instance, unauthorized construction monitoring concerns building changes, and flood disaster management addresses water changes. Most supervised networks utilize labels to guide the model to focus on the changes of interest. Obviously, pixel-level CD labeling is costly, so it is necessary to employ alternative technologies to extract universal change features.
2) *CD labels are difficult to guide semantic feature learning:* CD deep learning models often rely on binary labels, emphasizing changed versus unchanged features. However, they ignore the semantic information hidden in the CD dataset, limiting feature representation in complex scenarios.

To track these challenges, a new machine learning paradigm, self-supervised learning (SSL), has attracted considerable attention in RS [13], [14], [15], [16], [17], [18], [19], [20], [21]. These models are trained on RS images and provide domain-specific knowledge, enhancing interpretability for downstream tasks compared to random or ImageNet initialization. Although existing SSL methods have achieved remarkable success in representation learning for RS images, the discrepancies between pretraining and finetuning tasks limit their effectiveness for downstream tasks. Thus, some task-specific SSL methods have been explored, yielding improved finetuning results [22], [23]. However, specialized CD pretraining networks have seldom been explored [24], [25], [26]. Cai et al. [24] proposed a task-related SSL CD network using contrastive learning, which generates similar and dissimilar pair-wise samples based on image segmentation. However, segmentation quality affects sample selection, and besides, this method ignores the temporal information that is crucial for CD. Chen and Bruzzone [25] proposed a self-supervised CD approach, which uses contrastive loss

Received 10 March 2025; revised 21 April 2025 and 19 May 2025; accepted 7 June 2025. Date of publication 13 June 2025; date of current version 9 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61901439 and in part by the Key Laboratory of Target Cognition and Application Technology under Grant 2023-CXPT-LC-005. *(Corresponding author: Lei Ma.)*

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Digital Object Identifier 10.1109/TGRS.2025.3579416

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for feature alignment between multiview images. However, it treats each set of bitemporal images as positive samples regardless of changes, causing the projection layer to inadvertently align their representations, which reduces the model's sensitivity to distinguishing changed features. Chen et al. [26] incorporated semantic supervision into the SSL framework for building CD, using background swap augmentation and building labels to separate foreground and background in the embedding space. However, it cannot detect changes in other land categories.

In sum, the training process of existing SSL methods ignores the essential characteristics of CD, thus limiting their performance. Specifically, the methods are as follows.

1) Generative methods, represented by masked autoencoder (MAE) [27], randomly mask out over 75% of the input image and reconstruct the missing patches using a vision transformer, thereby capturing local spatial representations. However, these methods focus on learning the relationships within individual images independently and do not consider the relationships between multitemporal images. Therefore, the extracted features are not robust enough to imaging condition changes, resulting in pseudochanges when transferred to CD downstream tasks. Additionally, using transformers as encoders demands high computing resources due to numerous parameters.
2) Contrastive methods bring similar samples closer while separating dissimilar samples through the pretext task. These methods are most widely used in CD tasks since temporal information can be leveraged, making the encoded features more robust to diverse imaging conditions. However, these methods take multitemporal images as positive samples and bring their representations closer, regardless of whether changes have occurred, reducing model discrimination in downstream CD tasks. In addition, contrastive methods, due to the projection layer, are more inclined to capture global representations, unsuitable for pixel-level CD tasks.

To tackle the aforementioned challenges, in this work, we propose an SSL pretraining framework for RS image CD (SSLCD). SSLCD focuses on binary CD, leveraging the intrinsic structure of multitemporal data to learn discriminative representations of changes while also accounting for temporal invariant representations linked to irrelevant factors, such as illumination variations. Specifically, we take several pairs of multitemporal images and employ a three-branch encoder to extract features from the pre-phase image, the post-phase image, and the augmented views of the post-phase image. This encoder is trained using a generative pretraining strategy. We introduce a self/cross-reconstruction (SCR) module that extends masked image modeling (MIM) to multitemporal images by predicting missing parts of the pre-phase image using both the pre-phase and post-phase images in parallel. We restrict the similarity between the pre-phase image and both its self-reconstructed image and the cross-reconstructed image, and constrain reconstruction invariance across augmented views. This ensures that the feature learning process captures not only intra-image relationships but also inter-image dependencies, making the learned high-level contextual representations spatiotemporally invariant. Additionally, we employ an intrinsic structure learning (ISL) strategy to enable the encoder to capture shallow-level features. ISL leverages feature invariance across views and enforces change consistency between the pre-phase and post-phase images, as well as its augmented views, to learn discriminatory representations corresponding to changes while simultaneously mitigating noise. As a result, a competitive pretrained model for the CD task is generated using a small-scale dataset and limited computing resources.

The contribution of our work can be summarized as follows.

1) An SSL pretraining framework for RS image binary CD is proposed, which leverages the intrinsic structure of multitemporal data to learn discriminative representations of changes while also accounting for temporal invariant representations linked to irrelevant factors.
2) SSLCD learns CD-related representation with the ISL strategy and the SCR module. ISL achieves shallow-level feature learning by enforcing feature invariance and change consistency across temporal phases and augmented views, while SCR enables high-level contextual learning by extending MIM to multitemporal images.
3) Models pretrained using SSLCD are extensively evaluated on three CD datasets, and the results demonstrate that SSLCD outperforms existing RS pretraining methods and the state-of-the-art CD methods.

II. Related Works

A. CD Methods

As early as the 1960s, scholars have carried out research related to the RS image CD. Generally, traditional CD methods can be roughly divided into arithmetical-based algorithms [3], [4], transformation-based algorithms [5], [6], and classification-based algorithms [7]. However, traditional methods rely on expert domain knowledge, resulting in insufficient representation of extracted features, thus limiting the CD performance.

With the remarkable advancements of deep learning, it automatically learns representative features from a set of data, showing potential in the CD field. The most widely adopted structures are the encoder-decoder-based FCNs and its variants, where the encoder extracts multilayer features of bitemporal images and the decoder utilizes these multilevel features to recover spatial information and generate CD maps through classification. Typically, Daudt et al. [28] designed three CD architectures according to the input form of image data, and they are fully convolutional early fusion (FC-EF), fully convolutional Siamese-concatenation (FC-Siam-Conc), and fully convolutional Siamese-difference (FC-Siam-Di). Later, researchers employ dense connection [29], multilevel aggregation [30], [31], and attention mechanism [32], [33], [34] to enhance the feature extraction capability, thereby improving the CD performance.

Recently, with the remarkable advancements of deep learning, other state-of-the-art deep learning models, such as

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LSTM [8], GCN [9], transformer [10], [11], and diffusion model [12], are also applied to detect changes. Chen et al. [8] proposed the deep siamese convolutional multiple layers RNN (SiamCRNN), stacking LSTM units for spatial–spectral–temporal features. Wang et al. [9] proposed the coordinate space and feature interaction-GCN (CF-GCN), using diverse projections for coordinate and feature interaction graph convolutions. Wen et al. [12] used the denoising diffusion probabilistic model (DDPM) to generate CD maps, leveraging variational inference for change-aware features, enhancing sensitivity to subtle data differences.

However, the aforementioned supervised CD networks are trained using application-specific labels, which guide the model to concentrate on the changes of interest. Consequently, the model's portability is compromised, particularly when handling complex RS scenarios.

B. Self-Supervised Learning

1) SSL for RS Images: Most recently, SSL has raised considerable attention in RS, since it can learn task-independent representations without human annotation labels. Existing methods can be divided into two categories: generative and contrastive.

Generative methods have been a focus of increasing attention in machine learning research, as recent studies combine the advantages of both generative and discriminative methods [35]. Specifically, in the field of RS, MIM is widely used to predict missing parts of the input image. For example, Sun et al. [13] adopted the MAE method based on ViT and Swin Transformer architectures and trained an RS foundation model framework (RingMo) using 2 million unlabeled RS image dataset. Later, researchers improved the feature representation capability by utilizing contrastive learning [14], integrating spatio-temporal information [15], [16], [17] and leveraging multiscale information [18], [19]. Muhtar et al. [14] proposed a contrastive mask image distillation (CMID) SSL framework, which combines contrastive learning with MIM in a self-distillation way. Hong et al. [15] developed SpectralGPT, a novel 3-D generative pretrained transformer that utilizes 3-D token generation for spatial–spectral coupling. Cong et al. [16] presented SatMAE, which includes a temporal embedding along with independently masking image patches across time to leverage temporal information. Noman et al. [19] created SatMAE++, which performs multiscale pretraining and utilizes convolution-based upsampling blocks to reconstruct the image at higher scales. Additionally, Li et al. [36] proposed MA3E, which integrates RS angle factors into the MIM training architecture.

Contrastive methods learn both invariant and distinguishable features by bringing representations of semantically identical inputs closer and separating those of different images [37]. Therefore, contrastive methods often adopt a Siamese-like architecture design, in which semantically identical inputs are typically two augmented views of the same image. For example, Mañas et al. [20] proposed a seasonal contrast (SeCo) framework, which collects Sentinel-2 images from four seasons and employs spatially aligned images over time to construct temporal positive pairs, making the learned features more invariant under different imaging conditions. Kang et al. [21] proposed the spatially augmented momentum contrast (SauMoCo), using spatial augmentation and momentum update for contrastive learning, exploiting geospatial semantic similarities.

The above methods leverage multimodal and multitemporal images or geographical knowledge to ensure that the learned features are more specifically tailored to the target itself, i.e., exhibit greater invariance across various imaging conditions. Consequently, these models are more suitable for transferring to tasks such as classification, semantic segmentation, and disaster detection [38]. However, CD tasks focus not only on spatiotemporal invariant features but also on change information. Specifically, generative-based methods focus only on learning relationships within individual images and ignore inter-image relationships, while contrastive methods draw positive sample representations closer together regardless of whether changes have occurred. Consequently, the discrepancy between pretraining and finetuning tasks often limits the effectiveness of the aforementioned models when applied to CD downstream tasks.

2) SSL for CD: For CD pretraining networks, the contrastive methods are more widely used, since RS images acquired at different phases in the same region as positive samples and those of different regions as negative samples for learning temporal-invariant features [25], [39], [40], [41], [42], [43]. For example, Chen and Bruzzone [25] proposed a self-supervised CD approach based on multiview images, where the pretrained network consists of an encoder and a projection layer and uses contrastive loss for feature alignment between the multiview images. Zhao et al. [40] explore a self-supervised contrastive framework, which promotes the model to accurately capture spatial, structural, and semantic information through the domain adapter (DA) and the hierarchical contrastive head. Qu et al. [44] propose a novel network by fusing temporal and differential information for self-supervised contrastive learning CD (TD-SSCD), which aims to mine and learn correlations between bitemporal and differential images. For the problem that contrast learning tends to capture global representations, some studies [41], [42] achieve image-level and pixel-level instance discrimination through global and local contrastive learning headers. Although these methods have made some modifications to contrastive learning for CD tasks, they have yet to address the fundamental issue: they treat bitemporal images as positive samples and bring their representations closer together regardless of changes, thereby reducing the model's sensitivity to distinguishing changed features.

In addition, Alvarez et al. [45] propose a self-supervised conditional generative adversarial network (S2-cGAN), which is trained through input image reconstruction and adversarial learning to generate the distribution of unchanged samples, but methods based on GANs suffer from issues such as unstable training and monotonous generation results. Cai et al. [24] proposed a task-related SSL CD network, which generates similar and dissimilar pair-wise samples based on image segmentation, but this method ignores the temporal information

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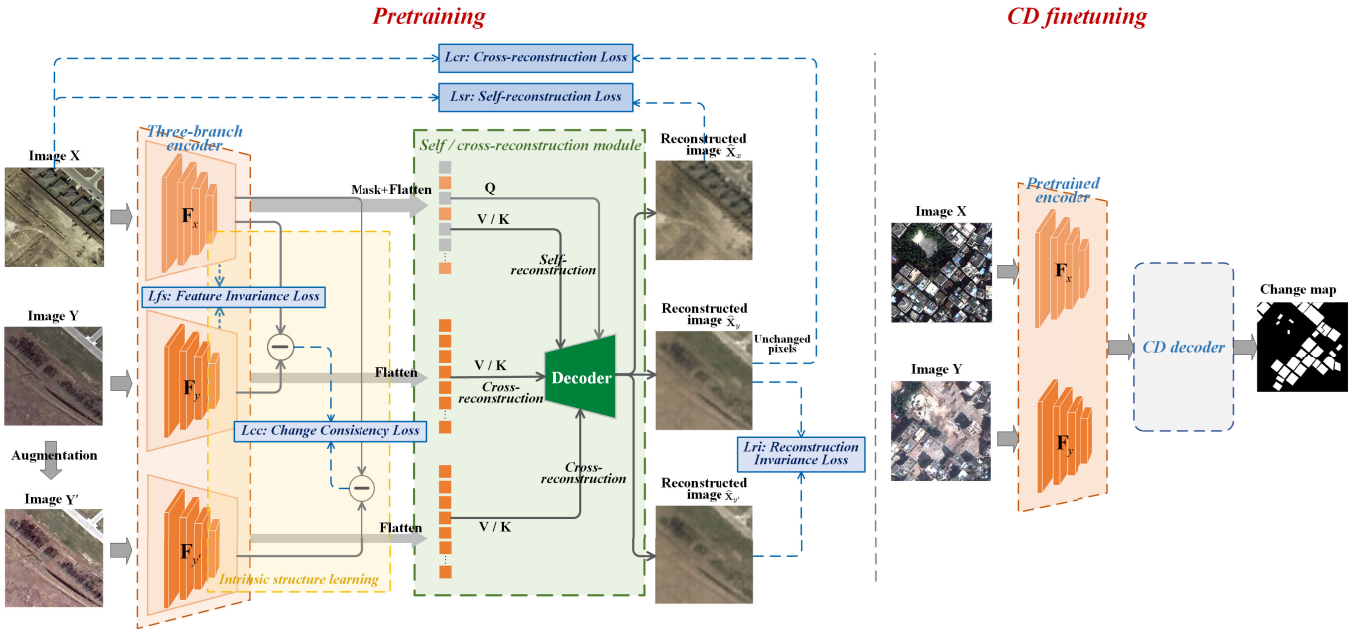


Fig. 1. Overview of the proposed SSLCD architecture. The encoder is trained utilizing the inherent information within the data, including shallow-level feature learning via the ISL strategy (manifested in the loss function L_{fs} and L_{cc}) and high-level contextual information learning through the SCR module (manifested in the loss function L_{cr}, L_{sr}, and L_{ri}).

that is crucial for CD. Wang et al. [46] propose a mutual prediction SSL (MPS2L), an MIM-based self-supervised method utilizing global, local, and temporal features for cross-temporal prediction, but its reconstruction process overlooks the impact of change information. Chen et al. [26] incorporated semantic supervision into the SSL framework and proposed a building CD method. They introduced a background swap augmentation and leveraged building supervision to learn an embedding space where background and foreground points are pushed apart. However, this method cannot detect changes in other land categories.

Therefore, specialized CD pretraining networks have rarely been explored, and the existing methods mainly concentrate on learning semantic representations or temporal-invariant features from multitemporal images, while neglecting the CD-related feature learning, thereby resulting in reduced sensitivity to distinguishing changes. Consequently, it is necessary to design a competitive pretrained model according to the task characteristics of CD.

In this work, we propose an SSLCD, which employs a generative pretraining strategy. Unlike existing methods, SSLCD extends MIM to multitemporal images by predicting missing parts of the pre-phase image using both the pre-phase and post-phase images in parallel. This ensures that the feature learning process not only focuses on the relationships within individual images but also on inter-image relationships, making the learned representations spatiotemporally invariant and robust to variations in imaging conditions.

III. Methodology

A. Overview

The overall structure of the proposed SSLCD pretraining framework is illustrated in Fig. 1. SSLCD starts with the three-branch encoder, which follows the Siamese structure and extracts features from the phase-1 image $\mathbf{X}$, phase-2 image $\mathbf{Y}$, and the augmented views of the phase-2 image $\mathbf{Y}'$, obtaining $\mathbf{F}_x$, $\mathbf{F}_y$, and $\mathbf{F}_{y'}$, respectively. Then, the feature maps are fed into the SCR module, which predicts the missing parts of $\mathbf{X}$ in parallel using $\mathbf{F}_x$, $\mathbf{F}_y$ and $\mathbf{F}_{y'}$, respectively.

The CD task-related representation is learned with ISL strategy and SCR module, which is specifically manifested in the loss function: L_{sr} constrains the similarities between image $\mathbf{X}$ and its self-reconstructed image $\widehat{\mathbf{X}}_x$ as the primary pretext task to guide the encoder learning; L_{cr} constrains the similarities between image $\mathbf{X}$ and its cross-reconstructed image $\widehat{\mathbf{X}}_y$; L_{ri} enforces reconstruction invariance across augmented views by making the cross-reconstructed image $\widehat{\mathbf{X}}_y$ and $\widehat{\mathbf{X}}_{y'}$ similar; L_{cc} constrains the change consistency between $\mathbf{F}_x - \mathbf{F}_y$ and $\mathbf{F}_x - \mathbf{F}_{y'}$; L_{fs} enforces feature invariance across temporal phases by bringing the feature distributions of $\mathbf{F}_x$ and $\mathbf{F}_y$ closer together.

B. Feature Extraction

In this research, we adopt ConvNeXt as the backbone to extract rich contextual information. ConvNeXt [47] is a standard CNN-based network that achieves comparable performance to the vision transformer network in terms of both accuracy and processing speed. We examine SSLCD using two models: ConvNeXt-T and ConvNeXt-A, and their configurations are shown in Table I.

Given an input image $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W represent the channels, height, width of $\mathbf{X}$, respectively. $\mathbf{X}$ undergoes a series of transformations starting with a convolution layer, followed by layer normalization, obtaining features with dimensions that are 1/4 of the input image size. Subsequently, these features pass through four stages, where each stage composed of a Downsample Layer and ConvNeXt Blocks, ultimately producing the multiscale feature

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TABLE I
DETAILED CONFIGURATIONS OF DIFFERENT CONVNEXT MODELS

Backbone	ConvNext-T		ConvNext-A	
stem	4×4, 96, stride 4		4×4, 40, stride 4	
res2	$\begin{bmatrix} \text{d7} \times 7, 96 \\ 1 \times 1, 384 \\ 1 \times 1, 96 \end{bmatrix}$	$\times 3$	$\begin{bmatrix} \text{d7} \times 7, 40 \\ 1 \times 1, 160 \\ 1 \times 1, 40 \end{bmatrix}$	$\times 2$
res3	$\begin{bmatrix} \text{d7} \times 7, 192 \\ 1 \times 1, 768 \\ 1 \times 1, 192 \end{bmatrix}$	$\times 3$	$\begin{bmatrix} \text{d7} \times 7, 80 \\ 1 \times 1, 320 \\ 1 \times 1, 80 \end{bmatrix}$	$\times 2$
res4	$\begin{bmatrix} \text{d7} \times 7, 384 \\ 1 \times 1, 1536 \\ 1 \times 1, 384 \end{bmatrix}$	$\times 9$	$\begin{bmatrix} \text{d7} \times 7, 160 \\ 1 \times 1, 640 \\ 1 \times 1, 160 \end{bmatrix}$	$\times 6$
res5	$\begin{bmatrix} \text{d7} \times 7, 768 \\ 1 \times 1, 3072 \\ 1 \times 1, 768 \end{bmatrix}$	$\times 3$	$\begin{bmatrix} \text{d7} \times 7, 320 \\ 1 \times 1, 1280 \\ 1 \times 1, 320 \end{bmatrix}$	$\times 2$

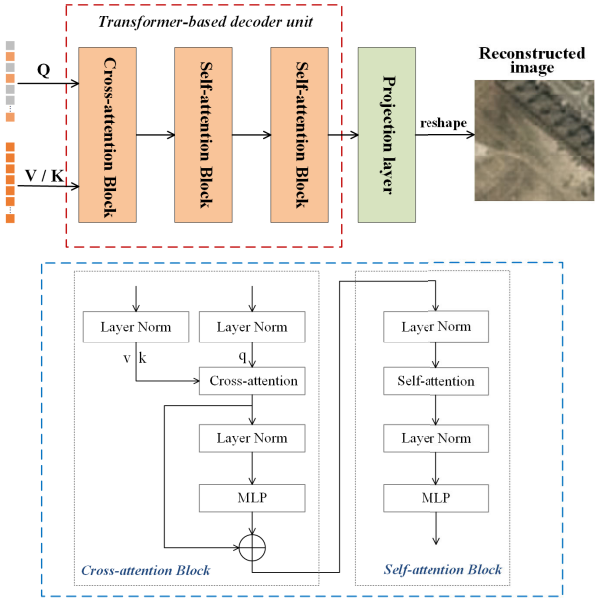


Fig. 2. Illustration of decoder architecture.

maps $\{\mathbf{F}_x^i | i = 1, 2, 3, 4\}$ with sizes that are 1/4, 1/8, 1/16, and 1/32 of the input size, respectively.

For each pair of multitemporal images, randomly select one temporal image as **X**, and the other as **Y**, and generate an augmented view **Y'** from **Y**. Then, the ConvNeXt-based encoder is utilized, sharing weights across all three branches for **X**, **Y**, and **Y'**, obtaining $\mathbf{F}_x$, $\mathbf{F}_y$, and $\mathbf{F}_{y'}$, respectively.

C. Self/Cross-Reconstruction

Reconstruction serves as the primary pretext task to guide the encoder learning. However, most reconstruction-guided methods concentrate on learning relationships within individual images independently and neglect inter-image relationships, which results in extracted features being non-robust to changes in imaging conditions. Therefore, we introduce an SCR module, which extends MIM to multitemporal images by predicting missing parts of the pre-phase image using the post-phase image, ensuring that the learned high-level contextual information remains spatiotemporally invariant.

1) Feature Masking: The feature map $\mathbf{F}_x$, extracted from **X**, is first partitioned into patches and then concatenated into a sequence of feature vectors $\mathbf{P}_x$ with a length of $(H/32) \times (W/32)$

$$\mathbf{P}_x = \text{flatten}(\mathbf{F}_x). \tag{1}$$

Similarly, $\mathbf{F}_y$ and $\mathbf{F}_{y'}$ are also converted to $\mathbf{P}_y$ and $\mathbf{P}_{y'}$ according to (1). Then, following the masking strategy employed in MAE [27], the sequence $\mathbf{P}_x$ is randomly sampled with a mask ratio of 90%. The sampled feature vectors are masked, and the remaining unsampled feature vectors form an unmasked sequence $M(\mathbf{P}_x)$.

2) Decoder: To reconstruct the original image **X**, a decoder is employed, which comprises a Transformer-based decoder unit and a projection layer, as depicted in Fig. 2.

The unmasked sequence $M(\mathbf{P}_x)$ is first projected using a linear layer, and then decoded jointly with a set of mask tokens, which serve as placeholders for the decoders to write the reconstructed patches.

The Transformer-based decoder unit integrates information from the encoded tokens, utilizing these tokens as queries and all the encoded tokens as keys and values, to recover feature sequences. Additionally, to enhance the capacity for learning high-level contextual information, in this research, we explore two decoder configurations: self-reconstruction and cross-reconstruction. Self-reconstruction takes $M(\mathbf{P}_x)$ as queries and $\mathbf{P}_x$ as keys/values, while cross-reconstruction utilizes $\mathbf{P}_y$ or $\mathbf{P}_y$ as keys/values. The images reconstructed with $\mathbf{P}_x$, $\mathbf{P}_y$, and $\mathbf{P}_{y'}$ are denoted as $\widehat{\mathbf{X}}_x$, $\widehat{\mathbf{X}}_y$, and $\widehat{\mathbf{X}}_{y'}$, respectively.

D. Loss Function

According to Fig. 1, the loss function is as follows:

$$L = L_{\text{sr}} + L_{\text{cr}} + L_{\text{ri}} + L_{\text{cc}} + L_{\text{fs}}. \tag{2}$$

Here, all terms in the loss function are considered equally because the scales of each component have been found to be on the same order of magnitude.

1) L_{sr} (Self-Reconstruction Loss): L_{sr} calculates the similarities between image **X** and its self-reconstructed image $\widehat{\mathbf{X}}_x$ using mean square error (mse), which is the primary pretext task to guide the encoder learning

$$L_{\text{sr}} = ||\widehat{\mathbf{X}}_x - \mathbf{X}||_2^2. \tag{3}$$

2) L_{cr} (Cross-Reconstruction Loss): L_{cr} calculates the distance between image **X** and its cross-reconstructed image $\widehat{\mathbf{X}}_y$ within the high-confidence unchanged regions. Here, we utilize the Euclidean distance between the multiscale encoded features $\mathbf{F}_x$ and $\mathbf{F}_y$ to generate a change intensity map. We then adopt a simple threshold strategy on this map, specifically, selecting the opposite minimum value of the standardized intensity map as the threshold [25], thereby identifying high-confidence unchanged regions.

The change intensity map **T** can be represented as follows:

$$\mathbf{T} = \text{Avg}\left[\mathbf{D}^1 + Up_{2x}(\mathbf{D}^2) + Up_{4x}(\mathbf{D}^3) + Up_{8x}(\mathbf{D}^4)\right] \tag{4}$$

where $\mathbf{D}^i = ||\mathbf{F}_x^i - \mathbf{F}_y^i||_2^2$; Up_{2x}, Up_{4x}, and Up_{8x} are bilinear upsampling functions with scale factors of 2, 4, and 8,

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respectively. Avg is the point-wise averaging operation used to aggregate multilevel change intensity maps. Additionally, here, we utilize four levels features $(\mathbf{F}_x^i, \mathbf{F}_y^i)$ to compute the difference map. This is based on our testing of various combinations of levels for constructing the difference map. We began by considering the lowest level, then incorporated the second-to-last level, and ultimately experimented with up to four levels. Through this progressive incorporation of levels, we discovered that the results obtained using four levels were the most optimal.

Then, the change map $\mathbf{M}$ is generated by classifying $\mathbf{T}$ using threshold ε, i.e., $\mathbf{M} = 0$ when $\mathbf{T} < \varepsilon$, indicating no change occurred. Thus, L_{cr} can be expressed as

$$L_{\text{cr}} = ||\widehat{\mathbf{X}}_y - \mathbf{X}||_2^2 \cdot \mathrm{I}_{[\mathbf{M}=0]} \tag{5}$$

where $\mathrm{I}_{[\cdot]}$ is the indicator function that returns 1 if the condition is met, and 0 otherwise.

3) L_{ri} (Reconstruction Invariance Loss): L_{ri} enforces similarity across augmented views during reconstruction, thereby making the encoded features more robust to diverse imaging conditions, especially in terms of contextual information

$$L_{\text{ri}} = ||\widehat{\mathbf{X}}_y - \widehat{\mathbf{X}}_{y'}||_2^2. \tag{6}$$

4) L_{fs} (Feature Invariance Loss): L_{fs} enforces feature invariance across temporal phases by calculating multiscale encoded features distance between $\mathbf{F}_x$ and $\mathbf{F}_y$ adopting symmetric Kullback–Leibler (KL) divergence

$$\begin{aligned} L_{\text{fs}} &= L_{\text{KL}}(P_{\mathbf{F}_y}||P_{\mathbf{F}_x}) + L_{\text{KL}}(P_{\mathbf{F}_x}||P_{\mathbf{F}_y}) \\ &= \sum_{i=1}^{N}\left[P_{\mathbf{F}_x}(i)\log\frac{P_{\mathbf{F}_x}(i)}{P_{\mathbf{F}_y}(i)} + P_{\mathbf{F}_y}(i)\log\frac{P_{\mathbf{F}_y}(i)}{P_{\mathbf{F}_x}(i)}\right]. \end{aligned} \tag{7}$$

5) L_{cc} (Change Consistency Loss): L_{cc} constrains change intensity map similarity between augmented views, enabling the model to learn discriminative representations of relevant changes while simultaneously reducing noises arising from irrelevant changes. It is calculated via symmetric KL divergence between $\mathbf{F}_x - \mathbf{F}_y$ and $\mathbf{F}_x - \mathbf{F}_{y'}$

$$\begin{aligned} L_{\text{cc}} &= L_{\text{KL}}(P_{\mathbf{T}_{x-y}}||P_{\mathbf{T}_{x-y'}}) + L_{\text{KL}}(P_{\mathbf{T}_{x-y'}}||P_{\mathbf{T}_{x-y}}) \\ &= \sum_{i=1}^{N}\left[p_{\mathbf{T}_{x-y'}}(i)\log\frac{p_{\mathbf{T}_{x-y'}}(i)}{p_{\mathbf{T}_{x-y}}(i)} + p_{\mathbf{T}_{x-y}}(i)\log\frac{p_{\mathbf{T}_{x-y}}(i)}{p_{\mathbf{T}_{x-y'}}(i)}\right] \end{aligned} \tag{8}$$

where $\mathbf{T}_{x-y}$ denotes the change intensity map calculated according to (4) using $\mathbf{D}^i = ||\mathbf{F}_x^i - \mathbf{F}_y^i||_2^2$, and $\mathbf{T}_{x-y'}$ is obtained similarly but with $\mathbf{D}^i = ||\mathbf{F}_x^i - \mathbf{F}_{y'}^i||_2^2$; N is the total number of pixels in the intensity change map, and here $N = H/4 \cdot W/4$.

E. Downstream CD Task

1) Decoder: We adopt a light CD decoder, which extends BIT [48] to a multiscale version, to generate the predicted change probability map $\mathbf{P}$. The structure of the CD decoder is illustrated in Fig. 3, and for more details, please refer to [48].

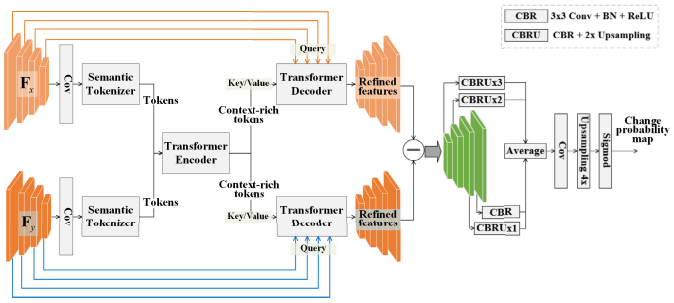


Fig. 3. Structure of CD decoder.

2) Loss Function: We follow our previous work [49], which is a hybrid loss that is composed of cross-entropy loss, contrastive loss, and Tversky loss

$$L_{CD} = L_{\text{ce}} + L_{\text{tver}} + L_{\text{contra}} \tag{9}$$

where (12), as shown at the bottom of the next page,

$$L_{\text{ce}} = -\frac{1}{N}\sum_{n=1}^{N}[G_n \log P_n + (1 - G_n)\log(1 - P_n)] \tag{10}$$

$$L_{\text{contra}} = \sum_{n=1}^{N}[\frac{1}{2}(1 - G_n)(P_n)^2 + \frac{1}{2}G_n\{\max(0, m - P_n)^2\} \tag{11}$$

L_{ce} is the cross-entropy loss, and G_n and P_n represent the label and predicted value for the nth pixel, respectively. L_{contra} is the contrastive loss. The changed pixels affect the loss function only when their predicted values fall within the margin m, and here, we set $m = 0.5$. L_{tver} represents the Tversky loss, while α is a hyperparameter that regulates the balance between recall and precision. In order to guarantee a high detection rate of the changed regions, we set $\alpha = 0.9$.

IV. Experimental Results

A. Pretraining Experimental Setup

1) Pretraining Dataset: We take 20 000 pairs of multitemporal images with 256 × 256 size from the S2Looking [50], SYSU-CD [51], and SECOND [52] datasets for SSLCD pretraining. Specifically, in order to create complex and comprehensive scenes suitable for practical applications, and considering the size of the SSLCD model, we randomly select 12 000, 4000, and 4000 image pairs from S2Looking, SECOND, and SYSU-CD datasets, respectively.

Datasets are described below.

S2Looking [50] contains 1 280 000 pairs (crop into 256 × 256 size) of side-looking images captured by various satellites, including GaoFen (GF), SuperView (SV), and BeiJing-2 (BJ-2). These images exhibit a high spatial resolution ranging from 0.5–0.8 m. Importantly, side-looking images are capable of capturing more details about objects, and thus the S2looking provides more complex and comprehensive scenes for practical applications. SYSU-CD [51] comprises 20 000 pairs of aerial images (256 × 256 size) with a resolution of 0.5 m, captured over Hong Kong. SYSU-CD focuses on change in high-rise buildings, and related changes around ports, as well as other significant changes like vegetation growth and road

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expansions. The SECOND [52] consists of 18 648 image pairs (crop into 256 × 256 size) with resolutions ranging from 0.5 to 3 m. These images were captured over cities such as Hangzhou, Chengdu, and Shanghai. The main types of changes include the transitions between land cover types such as water bodies, bare land, low-lying vegetation, trees, buildings, and sports fields.

2) Implementation Details: We pretrain the SSLCD network in the pretraining dataset without any labels. Here, we examine two versions of SSLCD, SSLCD-T, and SSLCD-A, with ConvNeXt-T and ConvNeXt-A as the backbone, respectively. The augmented views **Y'** are generated by combining the augmentation methods of color distortion, horizontal/vertical flipping, and Gaussian blur. The AdamW optimizer is employed to optimize the model, with a base learning rate of 0.0001, a momentum of 0.9, a weight decay of 0.05, betas of (0.9, 0.999), and a batch size of 64 for SSLCD-T and 256 for SSLCD-A. We warm up the model for 40 epochs and pretrain the model for 1000 epochs. The pretraining is conducted using one NVIDIA RTX A6000 GPU.

B. Downstream CD Experimental Setup

1) CD Datasets: We carried out experiments on three CD datasets.

1) *LEarning, VIsion and RS CD (LEVIR-CD):* The LEVIR-CD [32] is dedicated to building CD, particularly focusing on small and dense building change. It comprises 637 image pairs with 0.5 m resolution and 1024 × 1024 size. These images were sourced from Google Earth, spanning various cities in Texas, USA, and captured between 2002 and 2018. We adhere to its default dataset partition (training/validation/test) and crop the images into smaller, non-overlapping patches of 256 × 256 size. Consequently, we obtain 7120, 1024, and 2048 pairs of patches for training/validation/test.
2) *Change detection dataset (CDD):* The CDD [53] comprises 16 000 pairs of images with the size of 256 × 256. These images were captured by Google Earth and exhibit a very high spatial resolution ranging from 3 to 100 cm. The change information encompassed within the CDD includes buildings, roads, forests, cars, and tanks. We follow the default dataset split, deriving 10 000/3000/3000 pairs of patches for training/validation/test.
3) *High-resolution complex urban scene CD (HRCUS-CD):* The HRCUS-CD [54] comprises 11 388 pairs of high-resolution RS images with a size of 256 × 256 and a resolution of 0.5 m. These images were collected in Zhuhai, China, and contain two main acquisition areas. The first region primarily covers urban built-up areas, spanning a time from 2019 to 2022, with minor changes in building areas. The second region spans from 2010 to 2018, and contains a variety of complex scenes, including urban villages, industrial parks, cultural tourism facilities, vegetation disturbance, high-rise apartments, and other large contiguous buildings. We follow the default dataset split, deriving 7974/2276/1138 pairs of patches for training/validation/test.

2) Implementation Details: All networks are implemented using the PyTorch framework. The experiments are conducted on an NVIDIA RTX A6000 GPU with 24 GB of memory. We apply data augmentation to the training data, which includes horizontal and vertical flipping. The AdamW optimizer is employed to optimize the model, with a learning rate of 0.0003 and a momentum of 0.9. The training process runs for 200 epochs with a batch size of 16. After each epoch of training, validation is performed, and the model that performs best on the validation set is selected for evaluation on the test set.

3) Evaluation Metrics: The quantitative analysis employs three evaluation metrics to assess the performance of different methods. These metrics are recall, precision, and $F1$-score, which are defined as follows:

$$\begin{aligned} \text{precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F1 &= \frac{2\cdot\text{precision}\cdot\text{recall}}{\text{precision}+\text{recall}} \end{aligned} \tag{13}$$

where TP, FP, and FN are the number of true positives, false positives, and false negatives, respectively.

C. Comparison With Self-Supervised Methods

To evaluate the effectiveness of the proposed pretraining framework, we conduct a comparison with eight related methods, including two commonly used model initialization methods, two representative SSL methods, and four recently proposed RS SSL methods.

1) *Random initialization:* All the parameters of the networks are initialized by the zero-mean normal distribution with a deviation of 0.02. We use ConvNeXt-T as the backbone.
2) *ImageNet pretraining:* The backbone (ConvNeXt-T) of the CD model is initialized with the ImageNet pretraining.
3) *SimSiam [55]:* A representative SSL method directly maximizes the similarity between two augmented views of an image, using neither negative pairs nor a momentum encoder to prevent representation collapse. We use our pretraining dataset to train SimSiam, and subsequently finetune the downstream CD task using our CD decoder.

$$L_{\text{tver}} = 1 - \frac{\sum_{n=1}^{N} G_n P_n}{\sum_{n=1}^{N} G_n P_n + \alpha \sum_{n=1}^{N} G_n (1-P_n) + (1-\alpha) \sum_{n=1}^{N} (1-G_n) P_n} \tag{12}$$

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4) *MAE [27]:* MAE, a representative SSL method that reconstructs the missing patches with a vision transformer. We employ MViT [56], a multiscale transformer, as the backbone, and use our pretraining dataset to train MAE, and subsequently finetune the downstream CD task using our CD decoder.
5) *CMC-RSSR [57]:* SSL of RS scene representations using contrastive multiview coding, a recent SSL method in RS under the framework of contrastive multiview coding, which involves contrasting representations of different image views. We use our pretraining dataset to train CMC-RSSR, and subsequently finetune the downstream CD task using our CD decoder.
6) *SSLChange [40]:* A self-supervised CD framework based on domain adaptation, which promotes the model to accurately capture spatial, structural, and semantic information through the DA and the hierarchical contrastive head.
7) *MGC [58]:* An MLP-guided CNN pretraining method for RS images. MGC includes two encoders that first contrast pairwise samples and then apply a positive-pair guidance strategy to ensure consistency. Additionally, MGC introduces an attention guidance strategy to direct CNN branches toward foreground regions for effective discriminative learning. We use our pretraining dataset to train MGC, and subsequently finetune the downstream CD task using our CD decoder.
8) *RSP [59]:* An empirical study of RS pretraining, which trains different networks from scratch with the Million-AID dataset to obtain a series of pretrained backbones for RS, including both CNN and vision transformers such as Swin and ViTAE. We select the RSP-Swin-T version, leverage the RSP pretrained weights, and then finetune the model for the downstream CD task using our CD decoder.
9) *TOV [60]:* The original vision model for optical RS image understanding, a recent SSL approach in RS, introduces a human-mimicking learning mechanism. TOV first learns general knowledge from web-scale natural images and subsequently learns domain-relevant specialized knowledge from unlabeled RS images, achieving impressive generalization capabilities across various RS image understanding tasks. We use TOV pretrained weights, and then finetune the model for the downstream CD task using our CD decoder.

The SSL methods are implemented using their public codes with default hyperparameters.

1) Qualitative Analysis: Each pretrained model is evaluated by the performance of the downstream CD task. Fig. 4 provides the visualization comparison of different SSL methods. For a more intuitive comparison of the results, we use white to represent true positives, black for true negatives, red for false positives, and green for false negatives.

For the LEVIR-CD dataset, compared with the competing methods, SSLCD-T shows the best visual performance. Specifically, observing Fig. 4(a) and (b), the newly constructed buildings in the postchange image exhibit obvious spectral and shape characteristics, leading to generally accurate results by all methods. However, in comparison, our SSLCD-T has fewer false/misdetections than others. Additionally, SSLCD-A does not exhibit superior performance in these two sets of images, as its network is only 3.59M, and feature extraction is insufficient. Comprehensively observing Fig. 4(c) and (d), newly constructed buildings in the postchange image exhibit unconventional shapes in Fig. 4(c) and irregular arrangements in Fig. 4(d), with large shadows due to imaging conditions. All methods face false/misdetections, but the ImageNet/Random-pretrained methods are most severely impacted because they lack the utilization of prior knowledge related to RS, leading to significant false alarm areas in their results. In contrast, our SSLCD methods are the least affected by these factors and achieve the most satisfactory results. Observing Fig. 4(e), the newly constructed building in the postchange image is located in the shadow of the adjacent building, making it difficult to distinguish. Consequently, most methods fail to detect this change; however, our SSLCD-T successfully detects it.

For the CDD dataset, unlike the LEVIR-CD dataset, it also tracks changes caused by cars, trees, roads, etc., and we can see that our SSLCD-T achieves better results compared to others. To be specific, as shown in Fig. 4(f), the newly constructed building in the postchange image exhibits obvious spectral and shape differences compared to the surrounding environment. Consequently, all methods successfully detect changes, but our SSLCD-T achieves the most satisfactory result in terms of high internal compactness and minimal salt-and-pepper noises. Observing Fig. 4(g), all methods can monitor the new facility, but since the departing vehicle is too small, most methods fail to detect it. However, our SSLCD successfully detects this change, reflecting its capability in change information perception. In Fig. 4(h), seasonal factors cause disappearing buildings to resemble land. We can see that SimSiam, CMC-RSSR, and SSLChange methods fail to detect this change as they aim to align multitemporal/multiview images, reducing model discrimination in CD tasks. Other methods perform well, but SSLCD has fewer false detections and misdetections. As shown in Fig. 4(i), small car changes in multitemporal images cause most methods severe misdetections. However, the ImageNet-pretrained model is less affected. The reason may be that it is pretrained using the ImageNet dataset, which contains millions of images, including cars and multiscale objects. Additionally, our SSLCD has the fewest misdetections in comparison, reflecting a strong ability to capture changing characteristics. In Fig. 4(j), new roads have similar spectral characteristics to the surrounding environment, and the multitemporal images contain seasonal changes, leading to significant differences in the spectral representation. We can observe that the contrastive methods like SimSiam, CMC-RSSR, SSLChange, and MGC fail due to aligning multitemporal images and capturing global features, resulting in reduced discrimination for insignificant changes. SSLCD and MAE focus on semantic information, while the TOV method pays attention on category information, and thus they can successfully detect these changes. However, SSLCD-T achieves the most satisfactory

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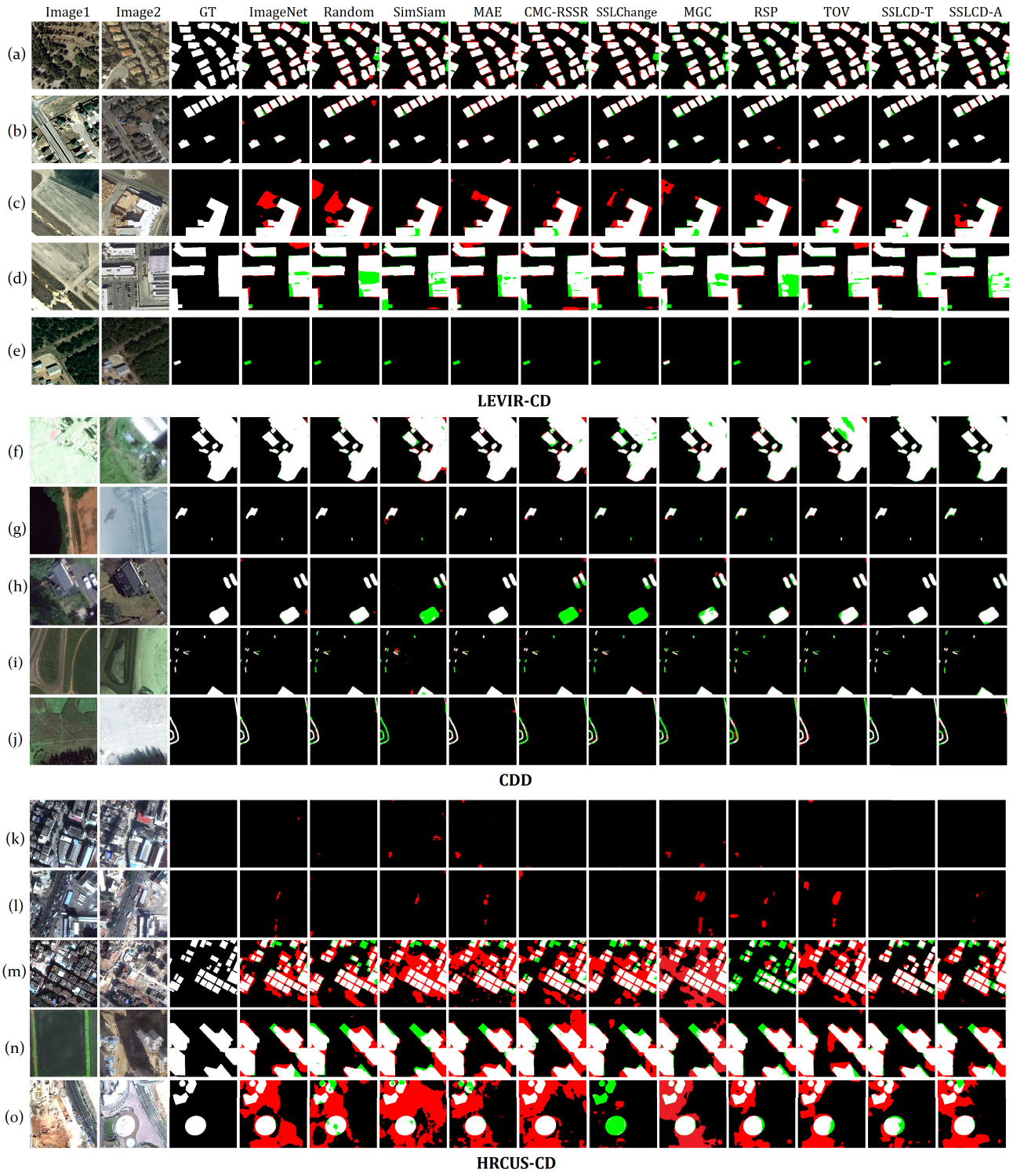


Fig. 4. CD results of different SSL methods. Legend: white for true positive, black for true negative, red for false positive, and green for false negative.

results, with minimal errors and a relatively clear detection edge.

For the HRCUS-CD dataset, which includes a variety of complex scenes, making the CD task significantly more challenging, we can observe that our SSLCD-T achieves more satisfactory results compared to others. For example, as shown in Fig. 4(k) and (l), despite no change between the bitemporal images, affected by the imaging conditions, noticeable geometric differences in tall buildings cause severe false alarms in the detection results. But, our SSLCD-T is less affected by this factor, benefiting from the SCR mechanism that enhances its ability to learn high-level contextual information. As shown in Fig. 4(m), the images contain dense buildings, shaded areas, and unrelated changes like heavy vehicle traffic, making CD challenging. In comparison, SSLCD exhibits the best visual performance with relatively fewer error detections, complete building structures, and high internal compactness, indicating our pretraining strategy effectively counters noise. Observ-

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TABLE II
QUANTITATIVE ANALYSIS OF DIFFERENT SSL METHODS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD					
			100%	20%	5%	1%
Method	backbone	Params.(M)/FLOPs(G)	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1
ImageNet	ConvNeXt-T	26.91 / 12.12	92.66 / 88.40 / 90.48	87.23 / 84.23 / 86.17	84.43 / 79.40 / 81.83	68.68 / 58.23 / 62.86
Random	ConvNeXt-T	26.91 / 12.12	91.40 / 85.89 / 88.56	82.01 / 79.81 / 80.89	72.43 / 66.37 / 69.25	58.75 / 38.14 / 46.16
SimSiam	Resnet-50	24.27 / 181.75	92.31 / 85.87 / 88.97	86.12 / 79.38 / 82.61	73.11 / 69.54 / 71.08	62.66 / 51.51 / 55.60
MAE	MViT	51.15 / 23.75	**93.23** / 84.76 / 88.79	85.26 / 78.37 / 81.67	68.81 / 68.94 / 68.40	19.12 / 20.99 / 16.99
CMC-RSSR	Resnet-50	48.15 / 21.97	91.17 / 87.60 / 89.35	84.93 / 81.19 / 83.01	**87.46** / 70.24 / 77.87	71.22 / 54.16 / 61.28
SSLchange	ResUNet-18	34.34 / 16.33	92.95 / 84.11 / 88.31	87.50 / 78.87 / 82.95	85.13 / 68.41 / 75.49	62.76 / 52.13 / 56.80
MGC	Resnet-50	24.01 / 11.36	85.61 / 86.51 / 86.06	66.20 / 81.14 / 72.89	64.09 / 76.87 / 69.89	55.77 / 60.07 / 57.61
RSP	Swin-T	28.20 / 125.88	90.80 / 89.53 / 90.16	86.90 / 85.41 / 86.15	84.19 / 81.58 / 82.85	68.15 / 59.25 / 63.38
TOV	Resnet-50	24.27 / 22.89	92.19 / 88.11 / 90.10	77.22 / 85.28 / 80.70	71.50 / 66.32 / 68.37	62.18 / 40.65 / 48.89
SSLCD-T	ConvNeXt-T	26.91 / 12.12	90.96 / **90.44** / **90.70**	**87.87** / **86.85** / **87.35**	85.96 / **83.14** / **84.53**	**74.38** / **64.55** / **68.90**
SSLCD-A	ConvNeXt-A	3.59 / 1.89	89.48 / 89.75 / 89.61	86.43 / 84.14 / 85.27	86.52 / 77.55 / 81.78	70.20 / 61.40 / 65.39
	CDD					
			100%	20%	5%	1%
Method	backbone	Params.(M)/FLOPs(G)	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1
ImageNet	ConvNeXt-T	26.91 / 12.12	**98.48** / 96.07 / 97.26	91.54 / 90.84 / 91.19	78.31 / 82.25 / 80.22	59.88 / 59.48 / 59.63
Random	ConvNeXt-T	26.91 / 12.12	97.27 / 95.18 / 96.21	87.70 / 85.19 / 86.40	78.04 / 66.14 / 71.58	58.65 / 47.62 / 52.56
SimSiam	Resnet-50	24.27 / 181.75	88.39 / 91.59 / 89.96	66.26 / 80.26 / 72.37	69.11 / 73.99 / 71.41	49.34 / 53.23 / 50.64
MAE	MViT	51.15 / 23.75	98.32 / 93.19 / 95.68	79.57 / 87.95 / 83.12	71.41 / 71.25 / 71.33	49.20 / 37.36 / 37.01
CMC-RSSR	Resnet-50	48.15 / 21.97	90.65 / 92.98 / 91.80	79.73 / 84.68 / 82.12	71.86 / 77.21 / 74.44	57.26 / 52.10 / 54.14
SSLchange	ResUNet-18	34.34 / 16.33	88.67 / 92.66 / 90.62	75.17 / 51.86 / 76.46	63.66 / 69.59 / 66.47	51.71 / 54.76 / 52.90
MGC	Resnet-50	24.01 / 11.36	92.58 / 93.84 / 93.21	81.47 / 81.75 / 81.57	62.82 / 74.36 / 68.03	52.41 / 53.73 / 52.31
RSP	Swin-T	28.20 / 125.88	93.91 / 94.70 / 94.30	89.53 / 90.04 / 89.77	77.19 / **84.28** / 80.58	53.27 / **72.21** / 61.28
TOV	Resnet-50	24.27 / 22.89	95.76 / 93.52 / 94.63	84.89 / 86.24 / 85.55	72.27 / 72.53 / 72.24	55.86 / 38.54 / 45.38
SSLCD-T	ConvNeXt-T	26.91 / 12.12	97.85 / **97.03** / **97.44**	**92.64** / **94.22** / **93.43**	**80.86** / 82.44 / **81.63**	**69.22** / 61.72 / **65.22**
SSLCD-A	ConvNeXt-A	3.59 / 1.89	96.21 / 95.61 / 95.91	89.48 / 90.22 / 89.85	77.85 / 78.38 / 78.07	63.14 / 58.90 / 60.93
	HRCUS-CD					
			100%	20%	5%	1%
Method	backbone	Params.(M)/FLOPs(G)	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1	Recall/Precision/F1
ImageNet	ConvNeXt-T	26.91 / 12.12	74.04 / 70.82 / 72.39	65.32 / 44.28 / 52.55	52.89 / 36.89 / 43.35	33.76 / 13.68 / 18.87
Random	ConvNeXt-T	26.91 / 12.12	65.71 / 56.44 / 60.72	62.53 / 43.59 / 51.23	46.31 / 26.95 / 33.90	41.68 / 15.73 / 22.57
SimSiam	Resnet-50	24.27 / 181.75	70.27 / 45.67 / 55.36	52.67 / 40.74 / 45.18	40.67 / 31.63 / 25.00	32.30 / 16.48 / 21.24
MAE	MViT	51.15 / 23.75	**76.12** / 51.66 / 61.55	57.24 / 43.93 / 49.52	37.79 / 31.90 / 34.57	35.41 / 2.14 / 3.99
CMC-RSSR	Resnet-50	48.15 / 21.97	70.29 / 62.11 / 65.95	66.50 / 40.86 / 50.04	46.30 / 32.42 / 37.77	29.72 / 15.07 / 19.98
SSLchange	ResUNet-18	34.34 / 16.33	70.92 / 47.02 / 56.55	**69.23** / 30.52 / 42.29	54.15 / 21.90 / 30.47	39.96 / 18.94 / 25.67
MGC	Resnet-50	24.01 / 11.36	57.62 / 56.75 / 57.18	48.06 / 42.60 / 44.74	35.73 / 36.86 / 36.20	29.71 / 21.40 / 24.88
RSP	Swin-T	28.20 / 125.88	72.03 / 72.48 / 72.25	60.35 / 54.87 / 57.09	44.24 / **43.93** / 43.56	**48.57** / 14.57 / 20.46
TOV	Resnet-50	24.27 / 22.89	**76.12** / 68.20 / 71.94	54.88 / 53.34 / 53.74	41.24 / 31.87 / 35.87	25.46 / 15.77 / 19.35
SSLCD-T	ConvNeXt-T	26.91 / 12.12	74.66 / **73.87** / **74.27**	63.73 / **55.77** / **59.47**	55.27 / 41.28 / **47.07**	41.19 / **29.35** / **34.20**
SSLCD-A	ConvNeXt-A	3.59 / 1.89	73.63 / 65.08 / 69.10	59.80 / 51.84 / 56.39	**57.04** / 37.36 / 45.10	37.91 / 25.83 / 30.73

ing Fig. 4(n), besides the newly constructed tall buildings, there are obvious seasonal variations between the bitemporal images, causing serious false alarms in shaded areas. But, our SSLCD-T has the fewest false alarms comparatively, demonstrating its robustness to appearance differences due to different imaging conditions. Observing Fig. 4(o), the newly constructed buildings in the postchange image exhibit unconventional shapes and irregular arrangements, along with unrelated changes, making CD challenging. Comparatively, our SSLCD-T exhibits fewer error detections, which relatively effectively overcomes the interference of pseudochanges.

In conclusion, SSLCD-T outperforms other SSL methods in CD, achieving minimal error detection and high internal compactness. This reflects that our pretraining strategy can reduce the pseudochanges influenced by imaging conditions, and pays attention to high-level contextual information. Additionally, compared to SSLCD-T, SSLCD-A does not exhibit superior performance in dealing with some complex scenes, as its network is only 3.59M, and feature extraction is insufficient.

2) Quantitative Analysis: Table II reports the quantitative comparison of our proposed method with other pretrained models on three test sets, with the optimal index bolded for emphasis. To evaluate the data-efficient performance, we establish several data regimes: 1%, 5%, 20%, and 100%, where each number represents the proportion of labeled data available in the original training set. For the 1%, 5%, and 20% regimes, in order to obtain more robust results, we randomly sample three sets of data in each proportion as training sets, train the model on each, and give the average of the three sets of CD test results.

As shown in Table II, we can see that on all three datasets, our SSLCD-T consistently outperforms other pretrained models in terms of the $F1$-score across various data regimes. Especially, SSLCD-T achieves 5.52%, 3.94%, and 8.53% higher $F1$-scores than the second-best pretrained methods in the 1% data regime for the LEVIR-CD, CDD, and HRCUS-CD datasets, respectively. Additionally, SSLCD-A is only 3.59M, making feature capture capability limited, and we can observe that its performance is comparable to other pretrained methods. However, when dealing with complex scenes, such as the HRCUS-CD datasets, SSLCD-A demonstrates superior performance in the 1% data regime compared to other pretrained methods.

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TABLE III
AUGMENTATION EXPERIMENT ON THREE DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD Recall/Precision/F1	CDD Recall/Precision/F1	HRCUS-CD Recall/Precision/F1
Base	90.13 / 88.72 / 89.42	**96.87** / 94.29 / 95.56	73.10 / 63.34 / 67.87
+Flip	89.97 / 89.38 / **89.67**	95.65 / **95.64** / 95.65	68.91 / **67.63** / 68.26
+Blur	89.69 / 88.88 / 89.29	95.57 / 95.60 / 95.58	**73.90** / 63.17 / 68.11
+Rotation	**91.16** / 87.11 / 89.09	95.41 / 95.17 / 95.29	70.66 / 65.10 / 67.77
+Noise	88.31 / 88.43 / 88.37	95.59 / 95.45 / 95.52	68.01 / 64.79 / 66.36
Final	89.48 / **89.75** / 89.61	96.21 / 95.61 / **95.91**	73.63 / 65.08 / **69.10**

TABLE IV
ABLATION STUDY ON THREE DATASETS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD Recall/Precision/F1	CDD Recall/Precision/F1	HRCUS-CD Recall/Precision/F1
C_1	**92.83** / 86.25 / 89.42	97.84 / 95.93 / 96.87	72.88 / 66.04 / 69.30
C_2	92.31 / 86.68 / 89.40	98.00 / 95.65 / 96.81	70.69 / 64.86 / 67.65
C_3	91.24 / 88.36 / 89.78	**98.46** / 95.82 / 97.12	**75.52** / 64.24 / 69.43
C_4	91.86 / 86.42 / 89.06	98.18 / 94.06 / 96.07	74.24 / 61.30 / 67.15
SSLCD-T	90.96 / **90.44** / **90.70**	97.85 / **97.03** / **97.44**	74.66 / **73.87** / **74.27**

3) Model Efficiency: Table II provides the number of parameters (Params.) and the floating-point operations per second (FLOPs) for each image, which are used to compare the model efficiency of different SSL methods. We can see that the MAE has the largest number of parameters, which limits its performance in small data regimes, especially when dealing with the HRCUS-CD dataset, which contains many complex scenes and where the CD fails in the 1% data regime. Among the other methods, SSLCD-T has a moderate number of parameters and relatively low FLOPs, while achieving the best $F1$-scores. Meanwhile, SSLCD-A has a very small number of parameters and the lowest FLOPs, delivering competitive performance. Especially when dealing with different data regimes, its detection ability remains relatively stable.

D. Augmentation Methods

To generate the augmented views of image $\mathbf{Y}$, we take color distortion as the baseline, and add horizontal/vertical flip (probability $= 0.7$), Gaussian blur (probability $= 0.3$), $90°/180°/270°$ rotation (probability $= 0.7$), Gaussian noise (probability $= 0.3$), respectively, to determine the final augmentation method. Since the feature similarity among different views is used as a constraint during training, the selection of augmentation methods should premise that the image's regional content remains unchanged.

We examine these augmentation methods with SSLCD-A, and their performance is shown in Table III. Combining the results from three datasets, we can see that the augmentation strategies of flip and Gaussian blur have a positive effect on the CD performance, but rotation and Gaussian noise have a negative effect. The reasons may be as follows: rotation augmentation forces the pretrained model to establish direction-independent feature expressions. However, MAE needs to understand the local and global structure during pretraining; rotation may make it difficult for the model to correctly capture these relationships, which will weaken the detection ability of downstream tasks in detecting direction-sensitive changes, such as new roads and building expansion. In contrast, flips preserve structural integrity by introducing mirror symmetry without distorting spatial relationships. Noise augmentation causes the pretrained model to excessively suppress high-frequency details during encoding, potentially destroying structural information in the image. This is particularly problematic in MAE, where the model must infer masked regions from surrounding data. This introduces irrelevant perturbations, leading the model to learn noisy patterns rather than genuine structural features. In contrast, Gaussian blur mitigates high-frequency noise (e.g., vegetation texture) and emphasizes low-frequency structures (e.g., building outlines, road networks), which are often pivotal for CD.

Thus, we ultimately select the augmentation method that combines color distortion, horizontal/vertical flipping, and Gaussian blur.

E. Ablation Studies

For a more comprehensive understanding of our proposed SSLCD-T, we present ablation studies in this section. According to Fig. 2, Combination 1 (C_1) provides the effect of the augmented branch by removing the change consistency and reconstruction invariance constraints from the SSLCD-T framework. Combination 2 (C_2) shows the impact of the SCR module by removing the cross-reconstruction process. Combination 3 (C_3) presents the effect of feature invariance constraint by removing L_{fs} from the pretraining process. Combination 4 (C_4) combines C_1, C_2, and C_3, with its pretraining process resembling the traditional MAE approach.

Table IV and Fig. 5 provide the quantitative performance and visual results of different ablation studies, respectively.

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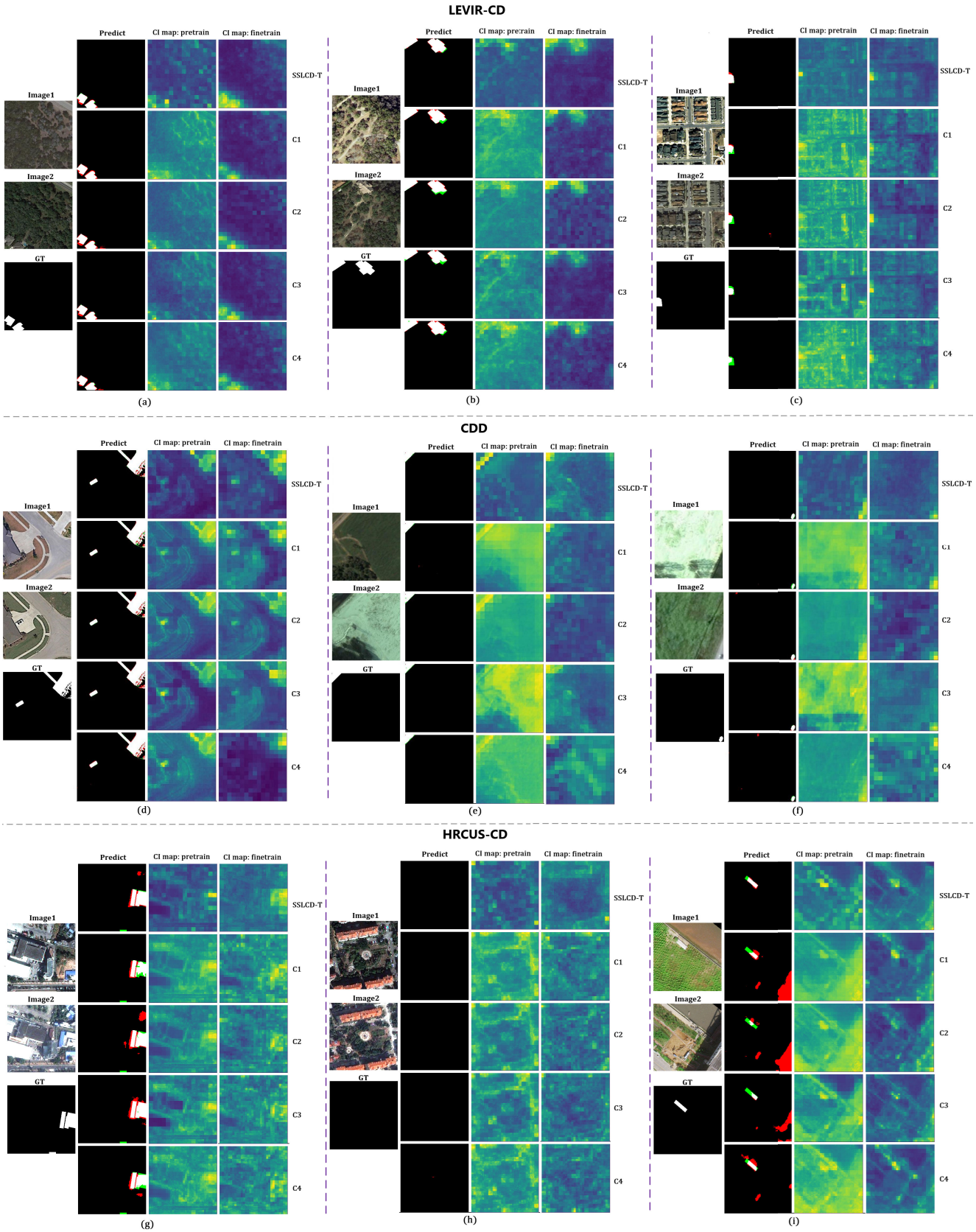


Fig. 5. Visual results of ablation studies. Predict: CD results of different ablation configurations are depicted with white indicating true positives, black for true negatives, red for false positives, and green for false negatives; CI map: pretrain and CI map: finetune: change intensity map $\mathbf{T}$ obtained using (4) during the pretraining phase and the finetuning phase, respectively.

1) Effect of the Augmented Branch: As shown in the results for C_1, we can observe that the augmented branch is critical in our SSLCD-T, as it introduces a dedicated branch specifically designed to consider the impact of imaging differences on

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TABLE V
QUANTITATIVE ANALYSIS OF DIFFERENT CD METHODS. ALL THE SCORES ARE DESCRIBED IN PERCENTAGE (%)

Method	Params.(M)	FLOPs(G)	LEVIR-CD Recall/Precision/F1	CDD Recall/Precision/F1	HRCUS-CD Recall/Precision/F1
FC-EF	1.29	3.57	83.85 / 80.10 / 81.93	75.57 / 85.10 / 80.06	23.01 / 52.43 / 31.99
FC-Siam-Di	1.29	4.72	85.54 / 84.89 / 85.22	78.99 / 86.85 / 82.73	45.98 / 38.63 / 41.99
FC-Siam-Conc	1.47	5.53	86.64 / 83.19 / 84.88	75.97 / 90.83 / 82.74	47.15 / 37.55 / 41.81
CDNet	16.19	71.58	90.63 / 81.14 / 85.62	91.22 / 90.07 / 90.64	73.08 / 62.29 / 67.26
BIT S4	3.39	8.71	89.37 / 89.24 / 89.31	91.73 / 95.69 / 93.66	65.43 / 69.67 / 67.49
ChangeFormer	39.13	202.79	88.80 / 92.05 / 90.40	93.94 / 94.53 / 94.23	64.04 / 68.85 / 66.36
CLDRNet	22.92	11.57	91.40 / 90.17 / **90.78**	96.16 / 95.64 / 95.90	**74.69** / 72.22 / 73.43
Dsfer-Net	32.51	73.05	**92.24** / 89.08 / 90.63	96.39 / 94.42 / 95.40	67.18 / **76.27** / 71.44
SEIFNet	27.28	8.37	88.77 / **92.27** / 90.49	96.32 / 96.25 / 96.29	60.17 / 73.93 / 66.34
SSLCD-T	26.91	12.12	90.96 / 90.44 / 90.70	**97.85** / **97.03** / **97.44**	74.66 / 73.87 / **74.27**
SSLCD-A	3.59	1.89	89.48 / 89.75 / 89.61	96.21 / 95.61 / 95.91	73.63 / 65.08 / 69.10

change-related feature learning. On the one hand, we enforce change consistency between image1 and image2 and its augmented views, making the extracted features related to changes and robust to imaging condition variations. On the other hand, we ensure that the reconstructed image1, obtained from these augmented views, remains similar, thus making the extracted features robust from the perspective of high-level context information. Consequently, the augmented branch can enhance the CD performance.

2) Effect of SCR Module: As shown in the results for C_2, the SCR module enhances the model's capacity in modeling high-level contextual information by predicting missing parts of image1 using the features of image2, thereby improving the CD performance. Additionally, by constraining high-level features, the SCR module implicitly constrains shallow-level features to a certain extent, addressing imaging variations. Therefore, C_2 demonstrates a more significant impact on CD performance compared to C_1 and C_3.

3) Effect of Feature Invariance Constraint: As shown in the results for C_3, the feature invariance constraint is crucial for CD. It brings the feature distributions of bitemporal images closer together, thereby making the extracted features robust to imaging differences. Despite the differences between bitemporal images, the changed areas usually only account for only a small part of the images, especially when dealing with a batch of images. Moreover, compared to constraints like mse, the KL distance makes their distribution close, rather than enforcing individual feature point values to be the same, making it more accommodating to changed pixels.

Additionally, observing the change intensity maps derived from different ablation configurations presented in Fig. 5, we can see that SSLCD can adapt to variations in imaging conditions and efficiently extract features associated with changes during the pretraining phase. During the finetuning phase, when guided by label information, the learned features become even more concentrated on changes of interest. This is particularly evident in the CDD dataset, which exhibits obvious seasonal variations, and the change intensity maps demonstrate marked differences between changed and unchanged areas. This illustrates the capability of SSTCD to address CD tasks with limited sample sizes.

Furthermore, as shown in Fig. 6, we use the LEVIR-CD dataset as an example and provide three sets of reconstruction results for different ablation configurations. We can observe that our SSLCD is capable of acquiring the content of Image 2 under the imaging conditions of Image 1, and conversely. Thus, SSLCD can reconstruct target information under specific imaging conditions, which reflects that the extracted features contain contextual information. The ablation configuration C_4 resembles MAE, which cannot learn the inter-information between bitemporal images. The reconstruction results of C_1 and C_3 are relatively unclear due to their lack of constraints on shallow-level features, which hinders the learning of detailed information. Observing the reconstruction results of C_2, we find that they poorly reproduce imaging conditions. Furthermore, comprehensively observing C_1, C_2, and C_3, we can see that the SCR module is crucial for extracting semantic information that is robust to imaging conditions, while the ISL strategy is important for understanding the details of the image.

F. Comparison With State-of-the-Art CD Methods

We make comparisons to several CD methods, including the purely convolutional-based methods: FC-EF [28], FC-Siam-Di [28], and FC-Siam-Conc [28], as well as related state-of-the-art research: CDNet [61], BIT [48], ChangeFormer [62], CLDRNet [49], Dsfer-Net [63], and SEIFNet [30]. We use their public codes with default hyperparameters.

Table V reports the quantitative comparisons of our proposed method with other CD models on three test sets, and we use recall, precision, and $F1$-score to evaluate their performances. We can see that on the multiclass CD datasets, including CDD and HRCUS-CD, our SSLCD-T significantly outperforms the competing methods in terms of $F1$-score. However, for the LEVIR-CD dataset, which is specifically dedicated to building CD, we notice that the performance of SSLCD-T does not demonstrate an advantage over other state-of-the-art CD methods. Because the pretraining of SSLCD is focused on learning discriminatory change-related

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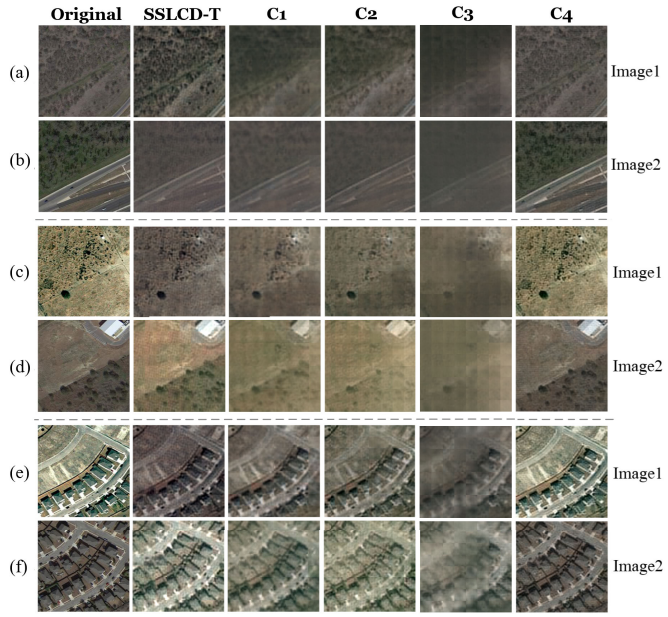


Fig. 6. Reconstruction results of different ablation configurations in the pretraining process. (a), (c), and (e) Reconstruction results for Image 1 utilizing the features of Image 2, while (b), (d), and (f) Reconstruction results for Image 2 employing the features of Image 1 (excluding the first column, which shows the original images).

representations with the intrinsic structure of multitemporal data, resulting in more generic features across different categories.

In addition, as shown in Table V, we use Params. and FLOPs for each image to compare the model efficiency of different methods. Comparing SSLCD-T to the state-of-the-art CD methods, SSLCD-T has a moderate number of parameters and relatively low FLOPs, while achieving relatively satisfactory $F1$-scores. Comparing SSLCD-A with the fully convolutional-based methods, including FC-EF, FC-Siam-Di, and FC-Siam-Conc, SSLCD-A has slightly more parameters and lower FLOPs, but the effect of CD is significantly improved.

V. Conclusion

In this work, we propose an SSLCD to generate competitive pretrained models for the CD task. SSLCD aims to learn general and robust change features by leveraging the intrinsic structure of multitemporal data. Specifically, SSLCD incorporates an ISL strategy and a SCR mechanism. It enforces feature invariance and change consistency across temporal phases and augmented views, learning discriminative representations related to changes, while simultaneously addressing temporal invariant representations associated with irrelevant changes. Finally, models pretrained using SSLCD are extensively evaluated on three CD datasets, and the results demonstrate that SSLCD outperforms existing RS pretraining methods as well as the state-of-the-art CD methods.

Additionally, our proposed framework focuses on binary CD, aiming to learn discriminative representations of changes. It implicitly captures high-level contextual information through a reconstruction pretext task. In the future, we plan to extend our pretraining framework to semantic CD tasks, for example, by incorporating a semantic contrast learning branch or leveraging class labels to enhance the model's awareness of distinct feature categories.

References

[1] L. Wan, M. Liu, F. Wang, T. Zhang, and H. J. You, "Automatic extraction of flood inundation areas from SAR images: A case study of jilin, China during the 2017 flood disaster," *Int. J. Remote Sens.*, vol. 40, no. 13, pp. 5050–5077, Jul. 2019.

[2] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989.

[3] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2007.

[4] F. Bovolo, S. Marchesi, and L. Bruzzone, "A framework for automatic and unsupervised detection of multiple changes in multitemporal images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 6, pp. 2196–2212, Jun. 2012.

[5] B. Parmentier, "Characterization of land transitions patterns from multivariate time series using seasonal trend analysis and principal component analysis," *Remote Sens.*, vol. 6, no. 12, pp. 12639–12665, Dec. 2014.

[6] L. Wanab, L. Maab, J. Guoad, M. Liuac, and D. Y. Ab, "Slow feature analysis based on convolutional neural network for SAR image change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3805–3808.

[7] L. Wan, Y. Xiang, and H. You, "An object-based hierarchical compound classification method for change detection in heterogeneous optical and SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9941–9959, Dec. 2019.

[8] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[9] W. Wang, C. Liu, G. Liu, and X. Wang, "CF-GCN: Graph convolutional network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.

[10] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[11] W. Liu, Y. Lin, W. Liu, Y. Yu, and J. Li, "An attention-based multiscale transformer network for remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 599–609, Aug. 2023.

[12] Y. Wen, X. Ma, X. Zhang, and M.-O. Pun, "GCD-DDPM: A generative change detection model based on difference-feature-guided DDPM," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–16, 2024.

[13] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612822.

[14] D. Muhtar, X. Zhang, P. Xiao, Z. Li, and F. Gu, "CMID: A unified self-supervised learning framework for remote sensing image understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023.

[15] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[16] Y. Cong et al., "SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 197–211.

[17] X. Li, D. Hong, and J. Chanussot, "S2MAE: A spatial–spectral pretraining foundation model for spectral remote sensing data," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27696–27705.

[18] C. J. Reed et al., "Scale-MAE: A scale-aware masked autoencoder for multiscale geospatial representation learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4088–4099.

[19] M. Noman, M. Naseer, H. Cholakkal, R. M. Anwer, S. Khan, and F. S. Khan, "Rethinking transformers pre-training for multi-spectral satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27811–27819.

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[20] O. Mañas, A. Lacoste, X. Giró-I-Nieto, D. Vazquez, and P. Rodriguez, "Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9414–9423.

[21] J. Kang, R. Fernandez-Beltran, P. Duan, S. Liu, and A. J. Plaza, "Deep unsupervised embedding for remotely sensed images based on spatially augmented momentum contrast," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2598–2610, Mar. 2021.

[22] H. Huang, Z. Mou, Y. Li, Q. Li, J. Chen, and H. Li, "Spatial–temporal invariant contrastive learning for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] S. Ma et al., "Unsupervised prototype-wise contrastive learning for domain adaptive semantic segmentation in remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.

[24] Z. Cai, Z. Jiang, and Y. Yuan, "Task-related self-supervised learning for remote sensing image change detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 1535–1539.

[25] Y. Chen and L. Bruzzone, "Self-supervised change detection in multiview remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.

[26] H. Chen, W. Li, S. Chen, and Z. Shi, "Semantic-aware dense representation learning for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[27] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 2022, no. June, pp. 15979–15988, Jun. 2022.

[28] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[29] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[30] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

[31] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–11, 2023.

[32] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[33] L. Wan, Y. Tian, W. Kang, and L. Ma, "D-TNet: Category-awareness based difference-threshold alternative learning network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[34] A. Chouhan, A. Sur, D. Chutia, and S. P. Aggarwal, "TMLNet: Triad multitask learning network for multiobjective based change detection," *Neurocomputing*, vol. 599, Sep. 2024, Art. no. 128080.

[35] P. Ren, Z. Han, Z. Yu, and B. Zhang, "Confucius tri-learning: A paradigm of learning from both good examples and bad examples," *Pattern Recognit.*, vol. 163, Jul. 2025, Art. no. 111481.

[36] Z. Li et al., "Masked angle-aware autoencoder for remote sensing images," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2024, pp. 260–278.

[37] P. Goyal, D. Mahajan, A. Gupta, and I. Misra, "Scaling and benchmarking self-supervised visual representation learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6391–6400.

[38] P. Li, Y. Wang, T. Si, K. Ullah, W. Han, and L. Wang, "MFFSP: Multiscale feature fusion scene parsing network for landslides detection based on high-resolution satellite images," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107337.

[39] U. Mall, B. Hariharan, and K. Bala, "Change-aware sampling and contrastive learning for satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5261–5270.

[40] Y. Zhao, T. Celik, N. Liu, F. Gao, and H.-C. Li, "SSLChange: A self-supervised change detection framework based on domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

[41] C. Wang, S. Du, W. Sun, and D. Fan, "Self-supervised learning for high-resolution remote sensing images change detection with variational information bottleneck," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5849–5866, 2023.

[42] F. Jiang, M. Gong, H. Zheng, T. Liu, M. Zhang, and J. Liu, "Self-supervised global–local contrastive learning for fine-grained change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.

[43] Y. Zhao, H.-C. Li, N. Liu, and R. Wang, "Toward distortion-aware change detection in realistic scenarios," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 10319–10322.

[44] Y. Qu, J. Li, X. Huang, and D. Wen, "TD-SSCD: A novel network by fusing temporal and differential information for self-supervised remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023.

[45] J. L. H. Alvarez, M. Ravanbakhsh, and B. Demir, "S2-CGAN: Self-supervised adversarial representation learning for binary change detection in multispectral images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 2515–2518.

[46] Q. Wang, Y. Qiu, P. Jin, T. Shen, and Y. Gu, "MPS2L: Mutual prediction self-supervised learning for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.

[47] S. Woo et al., "ConvNeXt v2: Co-designing and scaling ConvNets with masked autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 16133–16142.

[48] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 5, pp. 1–14, May 2021.

[49] L. Wan, Y. Tian, W. Kang, and L. Ma, "CLDRNet: A difference refinement network based on category context learning for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2133–2148, 2024.

[50] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021.

[51] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[52] K. Yang et al., "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609818.

[53] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLII-2, pp. 565–571, May 2018.

[54] J. Zhang et al., "AERNet: An attention-guided edge refinement network and a dataset for remote sensing building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023.

[55] X. Chen and K. He, "Exploring simple Siamese representation learning," in *Proc. IEEE Comput. Soc. Conf. Comput. Vision Pattern Recognit.*, Jun. 2021, pp. 15750–15758.

[56] Y. Li et al., "MViTv2: Improved multiscale vision transformers for classification and detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4804.

[57] V. Stojnic and V. Risojevic, "Self-supervised learning of remote sensing scene representations using contrastive multiview coding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1182–1191.

[58] Z. Li, B. Hou, W. Li, Z. Wu, B. Ren, and L. Jiao, "MGC: MLP-guided CNN pretraining using a small-scale dataset for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.

[59] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[60] C. Tao, J. Qi, G. Zhang, Q. Zhu, W. Lu, and H. Li, "TOV: The original vision model for optical remote sensing image understanding via self-supervised learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4916–4930, 2023.

[61] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403718.

[62] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[63] S. Chang, M. Kopp, P. Ghamisi, and B. Du, "Dsfer-net: A deep supervision and feature retrieval network for bitemporal change detection using modern Hopfield networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024.

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